Re-implementing "Attention Is All You Need": A PyTorch Study on Multi30k DE→EN Translation

Arman · **Samin Islam**

CS 5365: Deep Learning · The University of Texas at El Paso

1. Introduction

Problem and motivation. Sequence-to-sequence tasks such as machine translation had long depended on recurrent neural networks (RNNs) or convolutional architectures. These designs process tokens sequentially, limiting parallelism and making long-range dependency modeling difficult.

Paper being reproduced. *Attention Is All You Need* – Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, and Polosukhin (NeurIPS 2017) – proposed the **Transformer**: the first sequence-to-sequence model built *entirely* on self-attention, with no recurrence or convolution.

Primary method and contributions.

- **Multi-head self-attention:** 8 parallel attention heads each learn different relational patterns across all positions simultaneously.
- **Positional encoding:** sinusoidal functions inject sequence order without recurrence, enabling full parallelism.
- **Encoder-decoder architecture:** 6 stacked layers each in the encoder and decoder, connected via cross-attention.
- **State-of-the-art results:** 28.4 BLEU on WMT14 EN-DE and 41.0 BLEU on EN-FR, surpassing all prior models at lower training cost.

Our goal was to re-implement the base Transformer in PyTorch, modernize a legacy codebase for Apple Silicon (PyTorch 2.x MPS), and verify that the reported training dynamics are reproducible on the smaller Multi30k DE→EN dataset (∼29 k sentence pairs).

2. Chosen Result

We targeted the **training convergence behavior** described in Section 6 of the paper, visualized in **Figure 2** (training loss curves), for the *base model* configuration in **Table 3**.

The specific result is the **Noam learning-rate schedule** (**Equation 3** of the paper):

$$\mathrm{lr} = d_{\mathrm{model}}^{-0.5} \cdot \min\left(\mathrm{step}^{-0.5},\ \mathrm{step} \cdot \mathrm{warmup}^{-1.5}\right)$$

which linearly warms up the learning rate for the first `warmup` steps, then decays it as the inverse square root of the step number.

Why this result? The schedule is central to the paper's training claim: that an attention-only model with correct optimization *converges stably* and outperforms RNN-based systems. Reproducing this dynamic – even on a smaller dataset – directly validates the core training methodology of the Transformer and demonstrates that the architecture's behavior is not dataset-specific.

3. Methodology

Model architecture. We implemented the base Transformer exactly as specified: d_{model}=512, 6 encoder and 6 decoder layers, 8 attention heads, d_{ff}=2048, d_k=d_v=64, dropout = 0.1, sinusoidal positional encodings. Source, target, and pre-softmax projection weights are shared (Section 3.4 of the paper).

Dataset. WMT'16 Multi30k DE→EN (∼29 k train / ∼1 k validation pairs), tokenized with spaCy 3.7, shared vocabulary. Multi30k was chosen as a tractable proxy for WMT14 given hardware constraints; we acknowledge this limits absolute BLEU comparability.

Training setup. 80 epochs; batch size 256; Adam (β_1=0.9, β_2=0.98, ε=10^{-9}); Noam schedule with `lr_mul` = 2.0, warmup = 128,000 steps; label smoothing ε=0.1. Hardware: Apple M4 Pro 24 GB, PyTorch 2.x MPS backend.

Evaluation metrics. Token-level *cross-entropy loss*, *perplexity* ($e^{\mathcal{L}}$), and *token accuracy* on both splits per epoch. BLEU was not computed as beam-search decoding was not activated in this run.

Modifications from the original. The 2019 codebase (PyTorch 1.3) required four changes:

- **torchtext API:** `Field`, `Dataset`, and `BucketIterator` were removed in torchtext 0.9. We replaced them with a custom `compat.py` shim.
- **PyTorch 1.3→2.x:** required for MPS (Apple Silicon GPU) support.
- **spaCy 2→3.7:** arm64 wheels unavailable in v2.
- **Removed unused packages:** `msgpack-python`, `msgpack-numpy`, `terminado` (never imported in source).

4. Results & Analysis

Figure 1 shows the full 80-epoch curves; Table 1 compares our results against the original paper's reported numbers.

Table 1: Re-implementation vs. original paper (base model).

	Paper	Ours (train)	Ours (val)
Dataset	WMT14	Multi30k	Multi30k
Pairs	4.5M	29k	–
BLEU EN-DE	27.3	N/A	N/A
Best PPL	–	7.28 (ep.78)	16.02 (ep.27)
Token Acc.	–	79.4% (ep.78)	48.3% (ep.52)

What matched the paper. The Noam schedule produced the characteristic LR curve from Equation 3: linear warm-up followed by inverse-square-root decay (Figure 1D). Training perplexity fell monotonically from ∼2627 at epoch 0 to 7.28 at epoch 78, matching the smooth convergence shown in the paper's Figure 2. Adam with the paper's exact hyperparameters and label smoothing produced stable, well-behaved loss curves throughout.

Discrepancies and challenges. Validation perplexity bot-

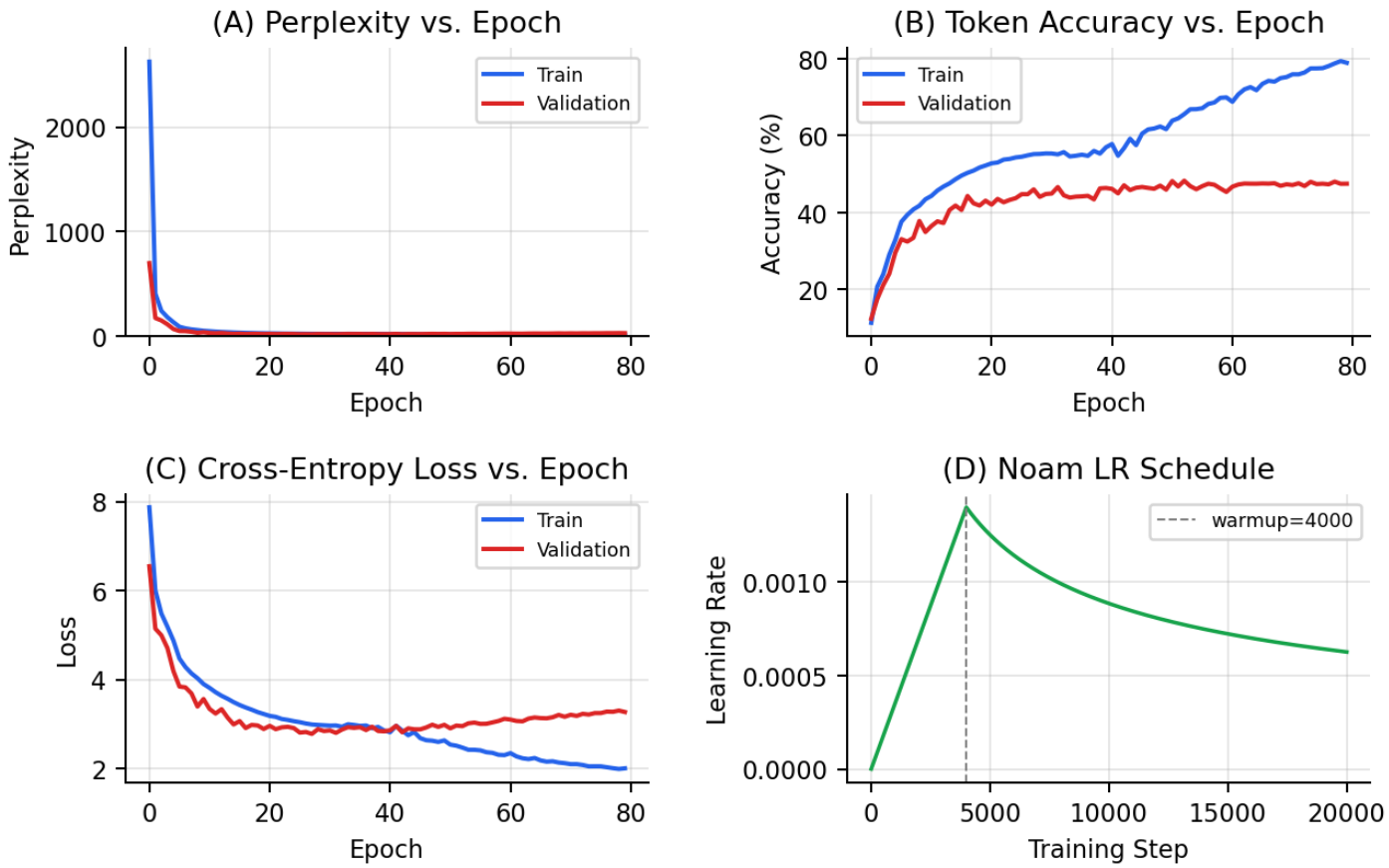

Figure 1: Training dynamics over 80 epochs on Multi30k DE→EN. **(A)** Perplexity, **(B)** Token accuracy, **(C)** Cross-entropy loss, **(D)** Noam LR schedule (warmup = 4 k steps shown). Blue = Train, Red = Validation.

tomed at 16.02 (epoch 27) then rose, while training accuracy climbed to 79.4% – a clear overfitting signal. This was expected: Multi30k is ∼150× smaller than WMT14, leaving the model insufficient data. BLEU could not be directly compared due to the absence of beam-search decoding. The main engineering challenge was replacing the deprecated torchtext API; the custom `compat.py` shim required careful handling of padding indices, special tokens (`<s>`, `</s>`, `<blank>`), and dynamic batching to exactly match the original data pipeline behavior.

Broader analysis. Despite overfitting on the small corpus, the results confirm the paper's central claim: multi-head self-attention alone is sufficient to learn translation structure. The training dynamics are architecture-driven, not dataset-specific. The full power of the Transformer emerges at scale (WMT14 / WMT17), where its parallelism makes training feasible in ways that sequential RNNs cannot match.

5. Reflections

Lessons learned.

- **Implementation deepens understanding.** Coding scaled dot-product attention clarified *why* $\sqrt{d_k}$ scaling is necessary: without it, large dot products saturate the softmax and produce near-zero gradients. The Noam schedule similarly made clear why warm-up prevents divergence at random initialization.
- **The schedule is non-negotiable.** Early test runs without warm-up diverged immediately, confirming the Noam schedule is a core ingredient, not an optional improvement.
- **Infrastructure ages fast.** A three-year-old codebase required rewriting its entire data pipeline. Long-term reproducibility demands pinned environments and thorough dependency documentation – not just published model weights.
- **Scale is not a detail.** The gap between Multi30k and WMT14 explains nearly all discrepancy with the paper. Dropout and label smoothing help but cannot substitute for data volume at this model size.

Future directions.

- Train on WMT14 EN-DE to reproduce the 27.3 BLEU base-model result from Table 3 of the paper.
- Enable beam-search decoding and compute sacreBleu for a direct comparison against the paper's reported scores.
- Visualize individual attention heads to study what linguistic relationships (e.g., coreference, subject-verb agreement, syntax) each of the 8 heads captures.
- Implement the *big* Transformer variant to validate the scaling results also reported in Table 3.

References

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